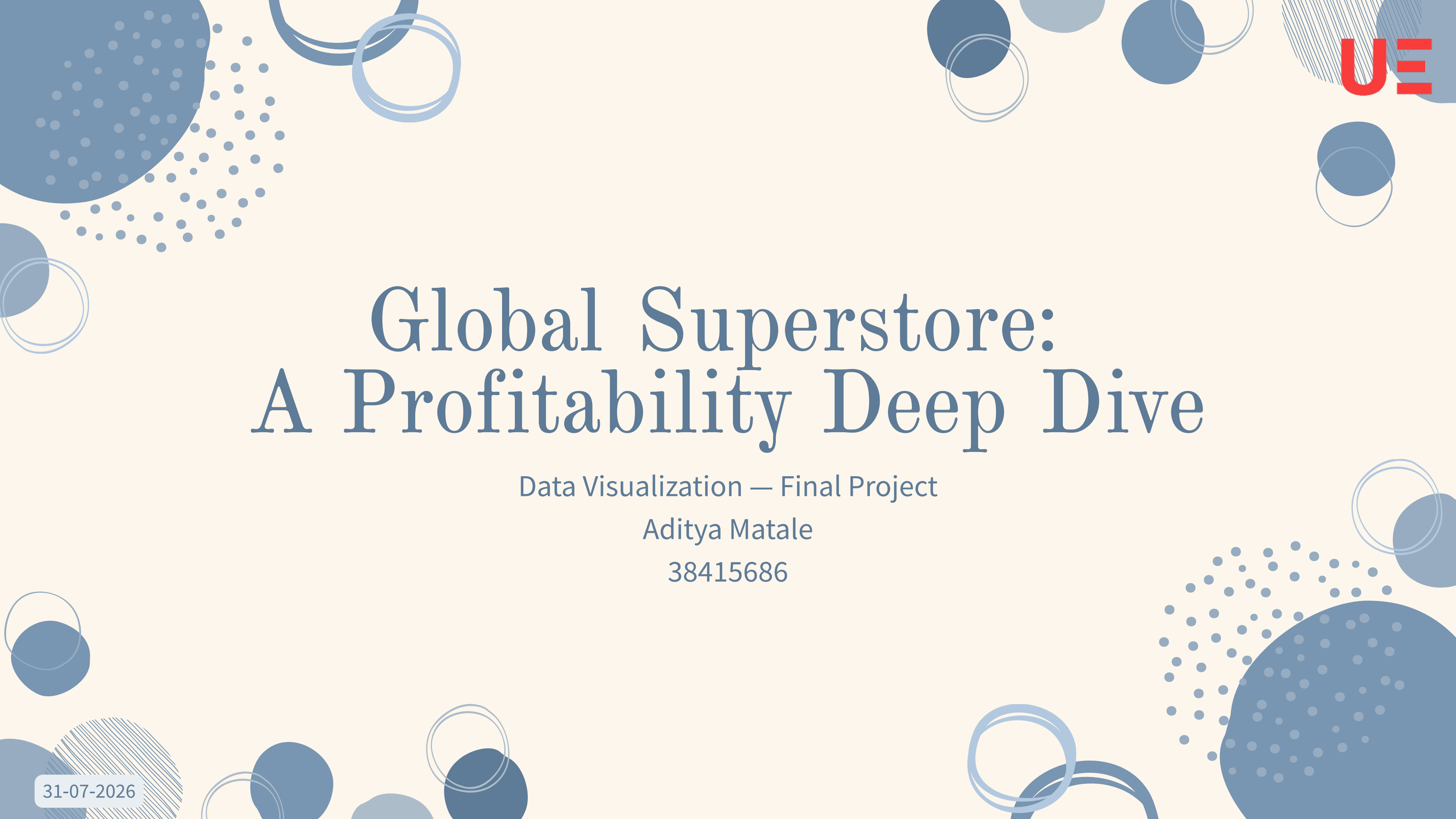




Global Superstore: A Profitability Deep Dive

Data Visualization — Final Project

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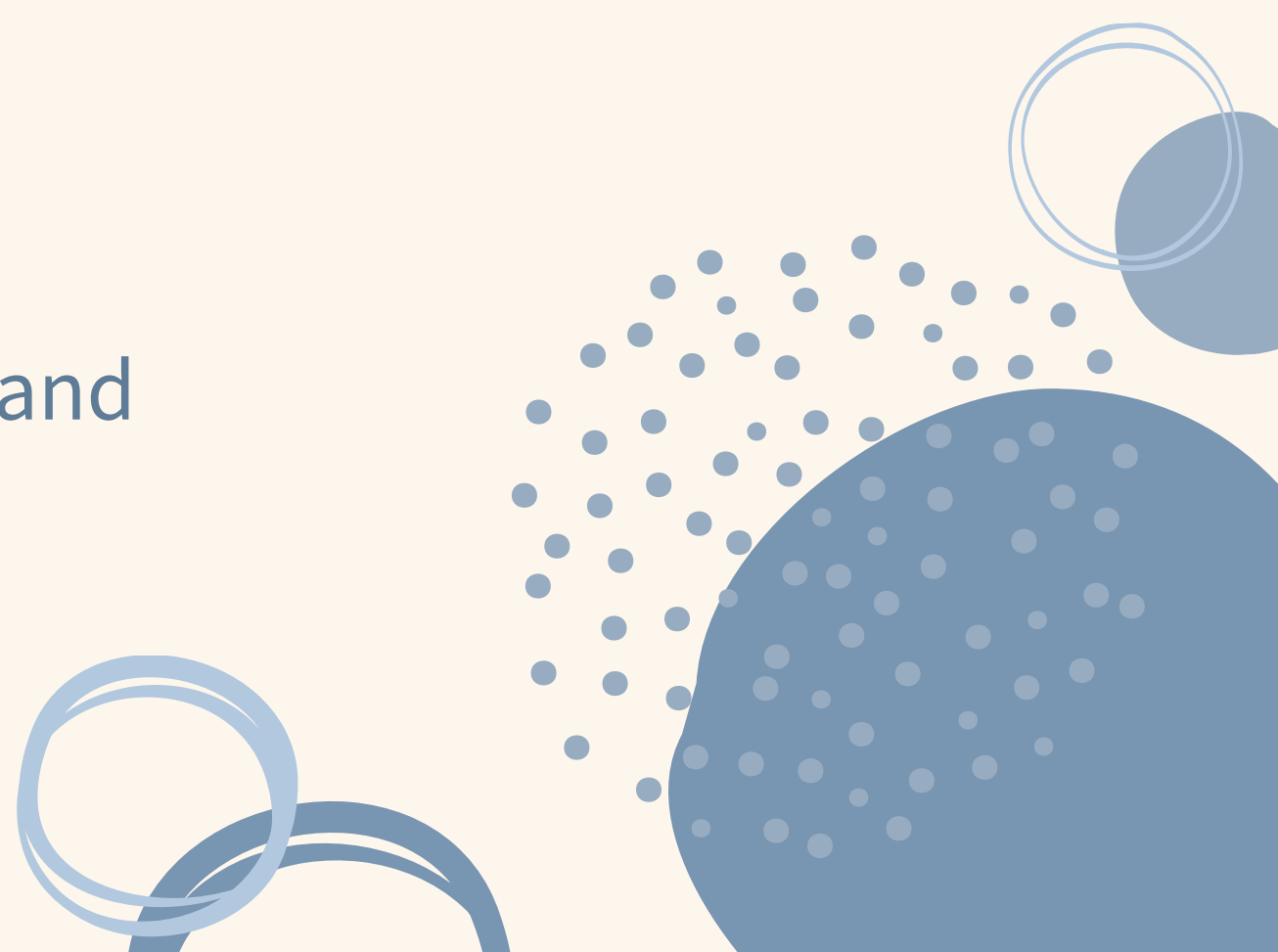
31-07-2026



Executive Summary

Global Superstore is profitable overall — but discounting practices and specific product lines are quietly eroding margin in predictable, fixable ways.

This analysis traces where profit is made, where it leaks, and where the real growth opportunity lies.



Dataset Card

Why this dataset:

Real-world retail data, rich across all four attribute types, supports genuinely multi-dimensional analysis.

[kaggle_dataset_url](#)

Field	Detail
Name	Global Superstore Dataset
Source	Kaggle
Scope	51,290 orders
Time Period	2011 – 2014
Geographic Coverage	147 countries, 7 global markets
Numerical Fields	Sales, Profit, Discount, Quantity, Shipping Cost
Categorical Fields	Category, Sub-Category (17), Segment, Ship Mode, Order Priority
Spatial Fields	Country, Market, Region, State, City
Temporal Fields	Year, weeknum
Missing Values	0
Duplicate Rows	0

Data Preparation

1. 0 nulls, 0 duplicate rows — clean at source
2. Dropped ji_lu-shu (constant column, no analytical value)
3. Order Date / Ship Date arrived corrupted
(placeholder text, no real dates)
 - Substituted with Year + weeknum → derived Period field
 - Yearly granularity still fully supports all 10 questions asked



"Where Is Profit Being Made — and Lost?"

(Act 1: context)

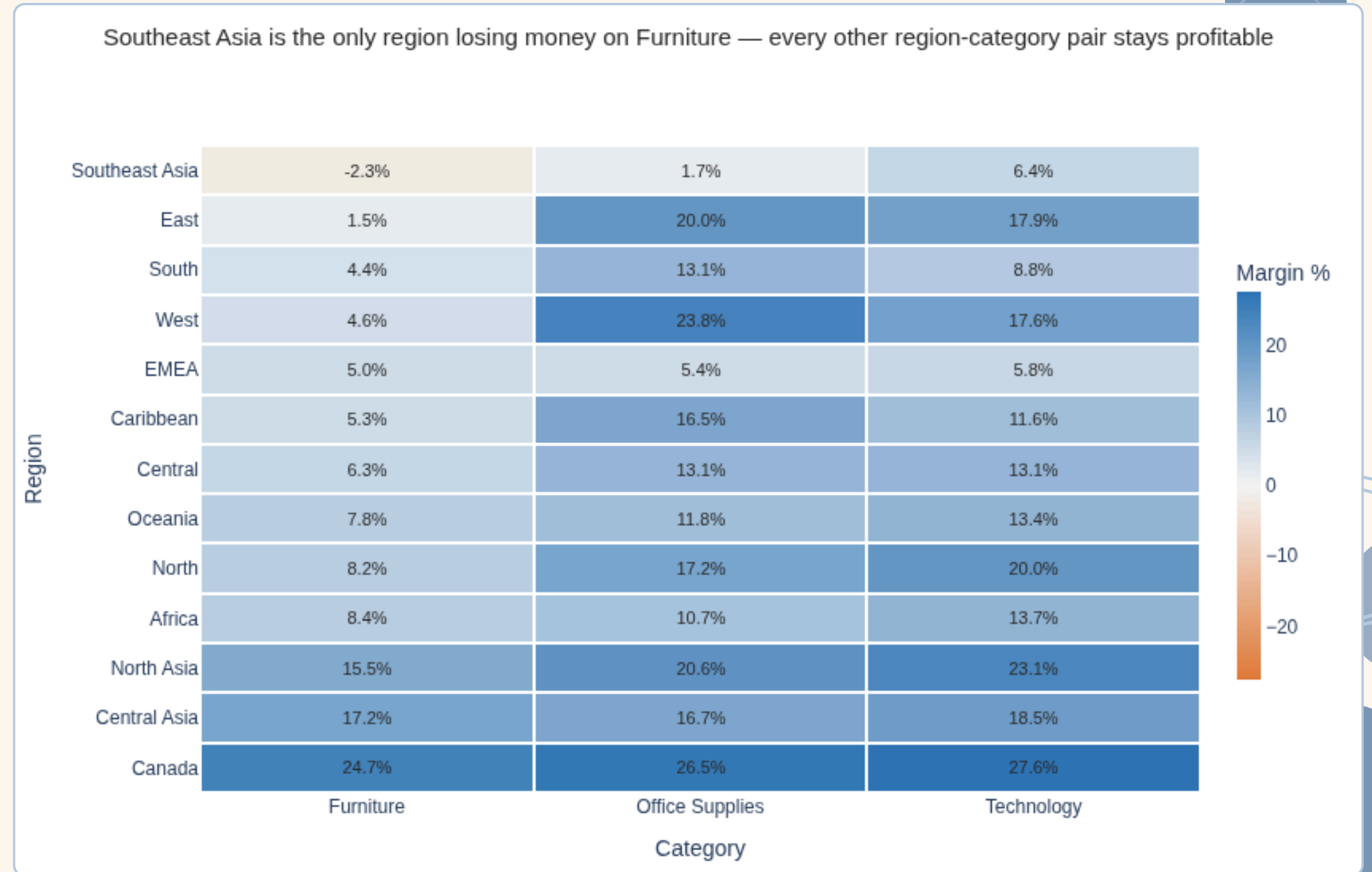
Which region–category combinations are quietly losing money despite high sales?

Chart type:

Heatmap — compares margin across two categorical axes (Region × Category) simultaneously.

Insight:

The loss isn't systemic — it's isolated to one region-category combination, making it a targeted fix rather than a structural problem.



Geographically, which countries drive the most profit — and which drive the most loss despite operating at real sales volume?

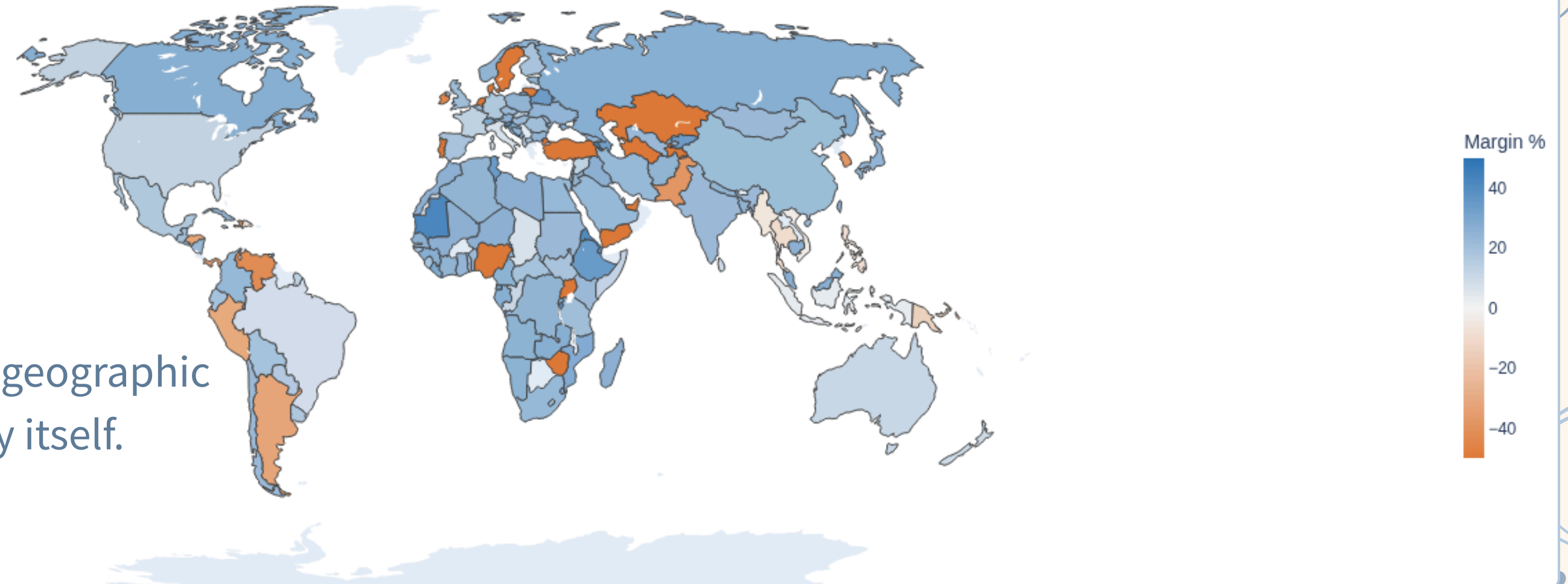
Nigeria loses money despite \$54K in sales — while major markets like China and India post the strongest margins (21%+)

Chart type:

Choropleth — the geographic pattern is the story itself.

Insight:

Scale and profitability aren't correlated — some of our largest markets are also our healthiest margins, while a mid-size market bleeds cash.



Does a faster/costlier ship mode actually correlate with higher order value or profitability?

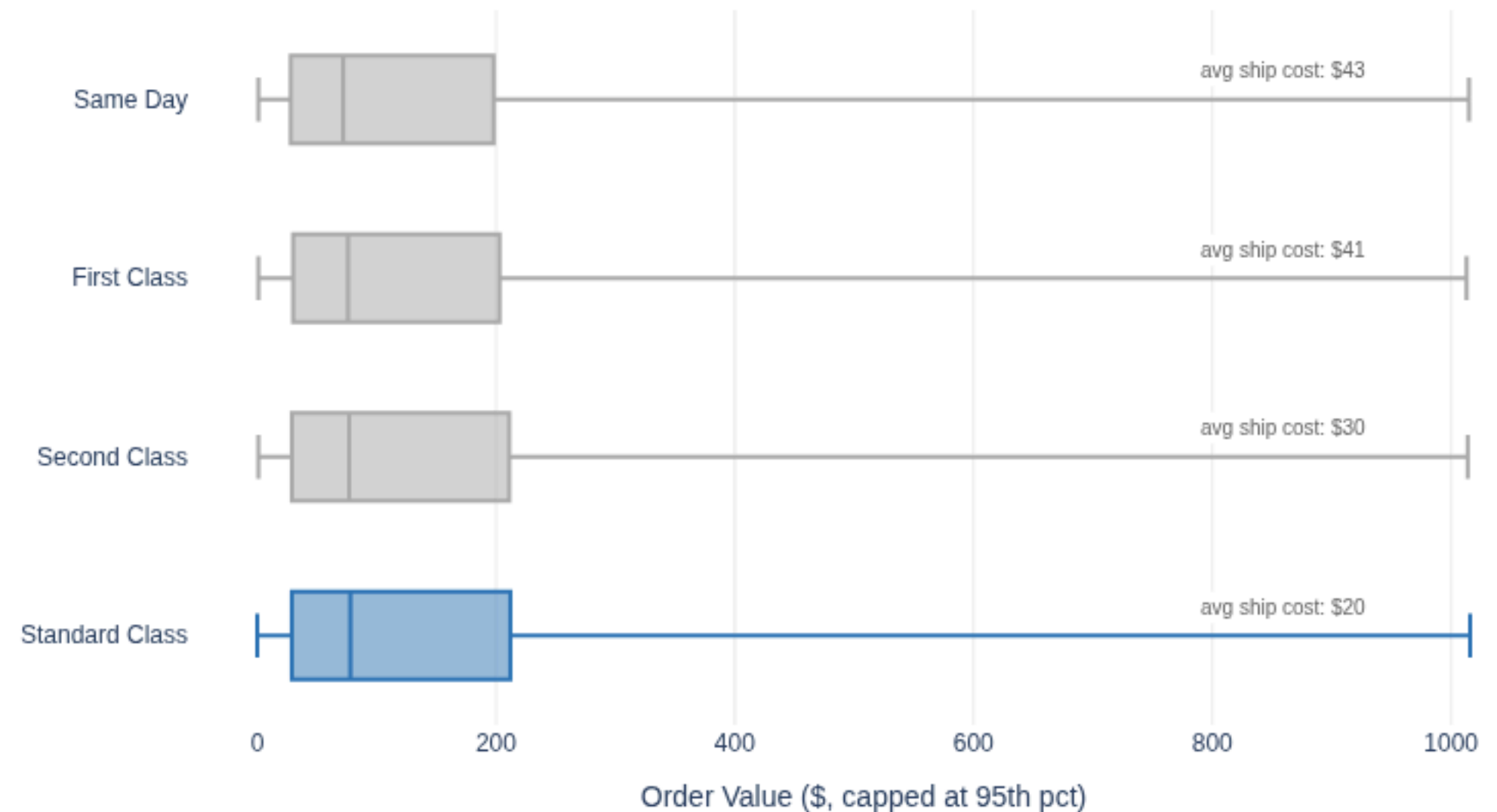
Chart type:

Box plot — compares full distribution, not just averages, across 4 ship modes

Insight:

Speed is a cost center, not a revenue driver — same-day shipping costs 2x more to fulfill (\$43 vs \$20) without customers spending any more.

Ship mode doesn't predict order value — customers who pay for Same Day spend the same as Standard Class





"Why Is Margin Eroding?"

(Act 2: The problem)

Does discount level affect profit margin across categories, and is there a tipping point where discounts destroy profit?

Chart type:

Bubble chart — Discount (x) vs Margin (y) vs Sales (size), coloured by Category

Reference:

Dashed breakeven line at 0%

Insight:

Margin declines steadily with discount; Several sub-categories cross into negative margin — Tables being the clearest tipping point.



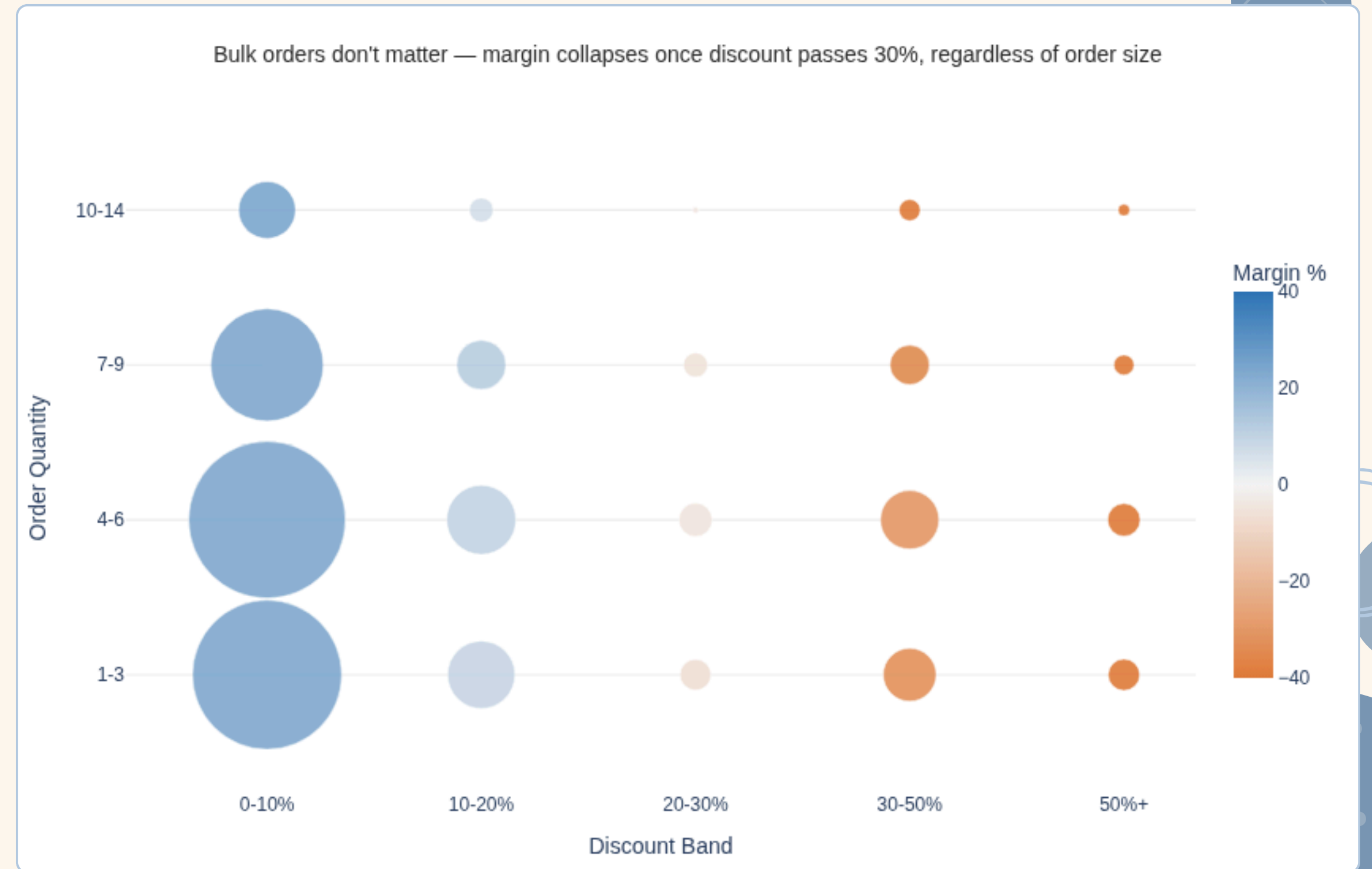
How does order quantity interact with discount to affect profitability — are big discounted bulk orders actually worth it?

Chart type:

Bubble matrix — Discount Band × Quantity Band grid, coloured by margin, sized by sales volume

Insight:

Quantity has almost no effect on margin — discount is the sole lever driving profitability down, at any order size.



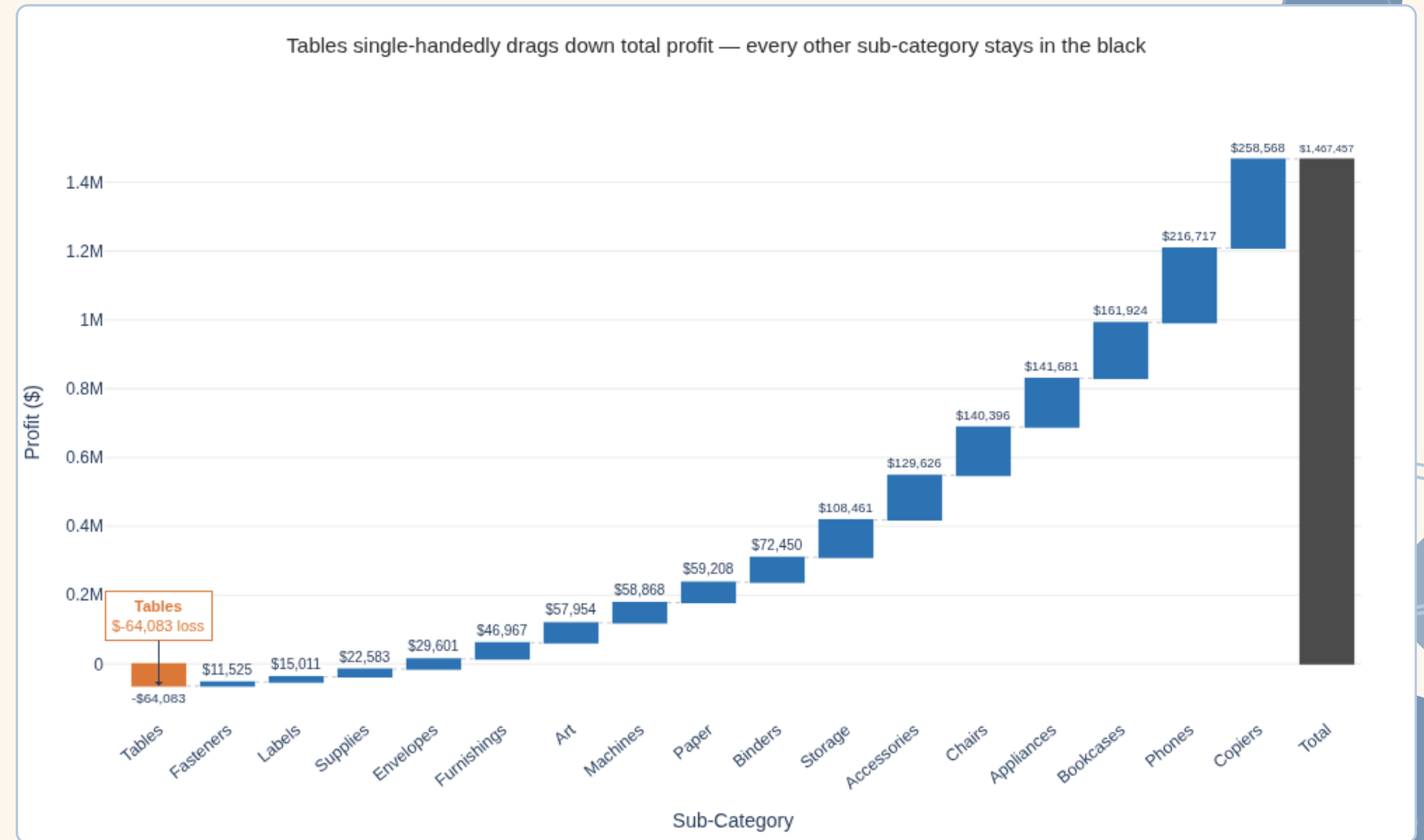
Which sub-categories contribute disproportionately to profit vs. loss within each category?

Chart type:

Waterfall — shows cumulative profit build-up sub-category by sub-category

Insight:

This isolates the problem to one specific product line (-\$64,083), not an entire category — confirms the fix is targeted, not broad.



"Where Is the Real Growth?"

(Act 3: The opportunity)

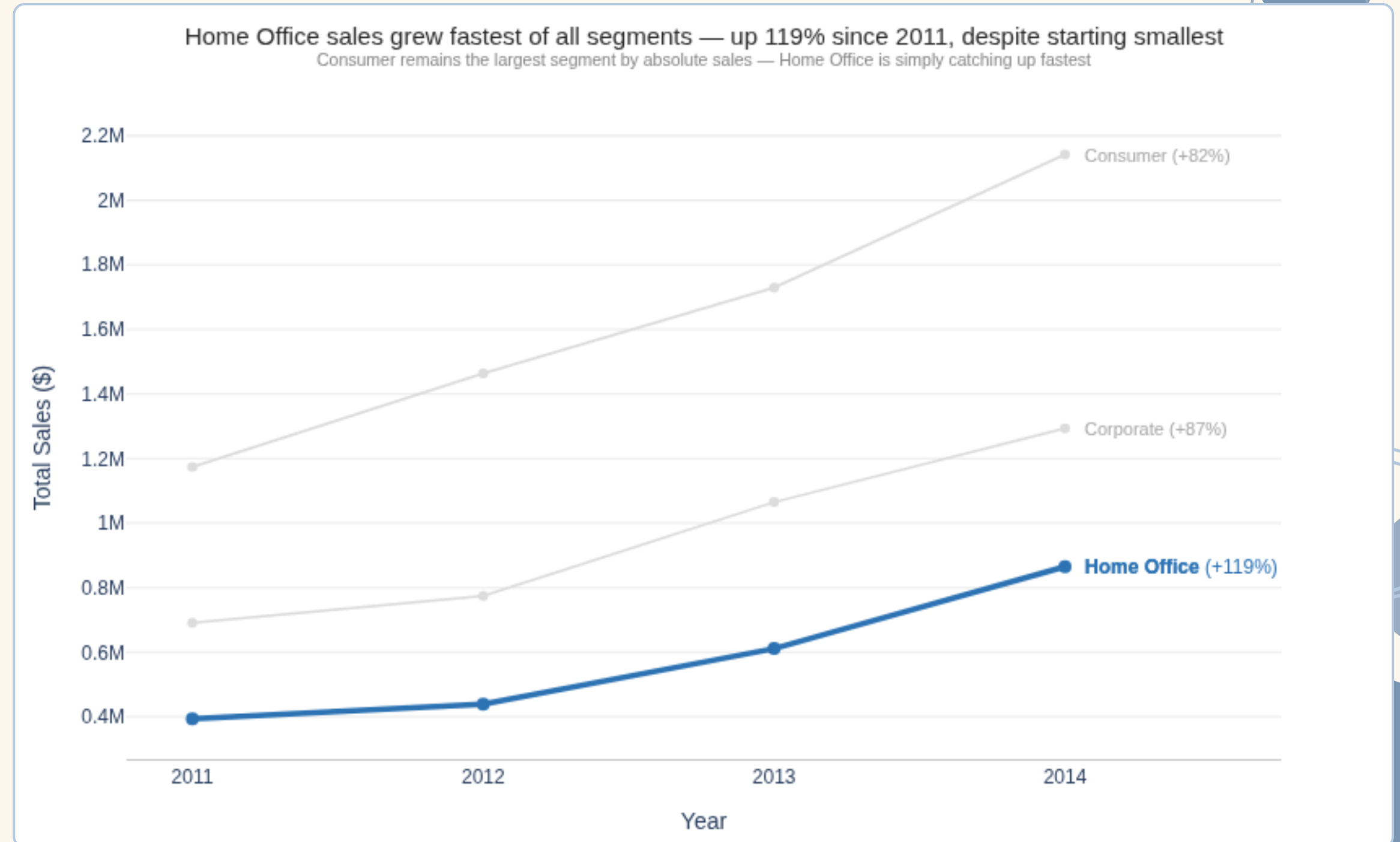
How have sales and profit trended over time across customer segments — did any segment's growth stall or reverse?

Chart type:

Multi-line with highlight — Year is a genuine trend

Insight:

Two true stories coexist — Consumer leads in scale, Home Office leads in momentum. Worth watching as a rising segment.



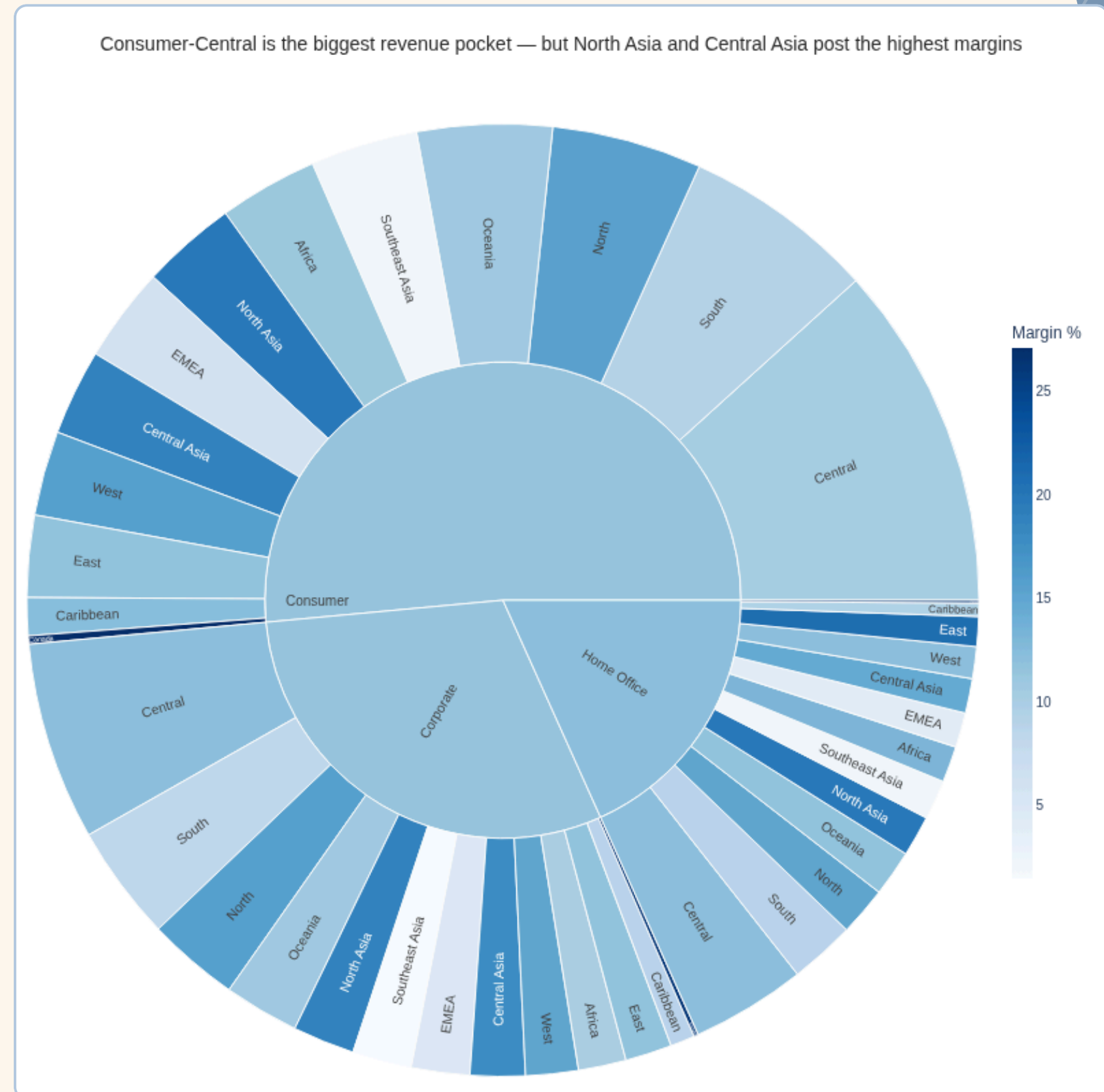
How does customer segment x region combination differ in average order value and profit margin?

Chart type:

Sunburst — Segment → Region hierarchy, sized by sales, coloured by margin (sequential, since all margins here are positive)

Insight:

Big spend and high margin don't always overlap — worth distinguishing "where the money is" from "where the best margins are."



Within the most profitable category, which sub-category's profitability changed most dramatically between the first and last year of data?

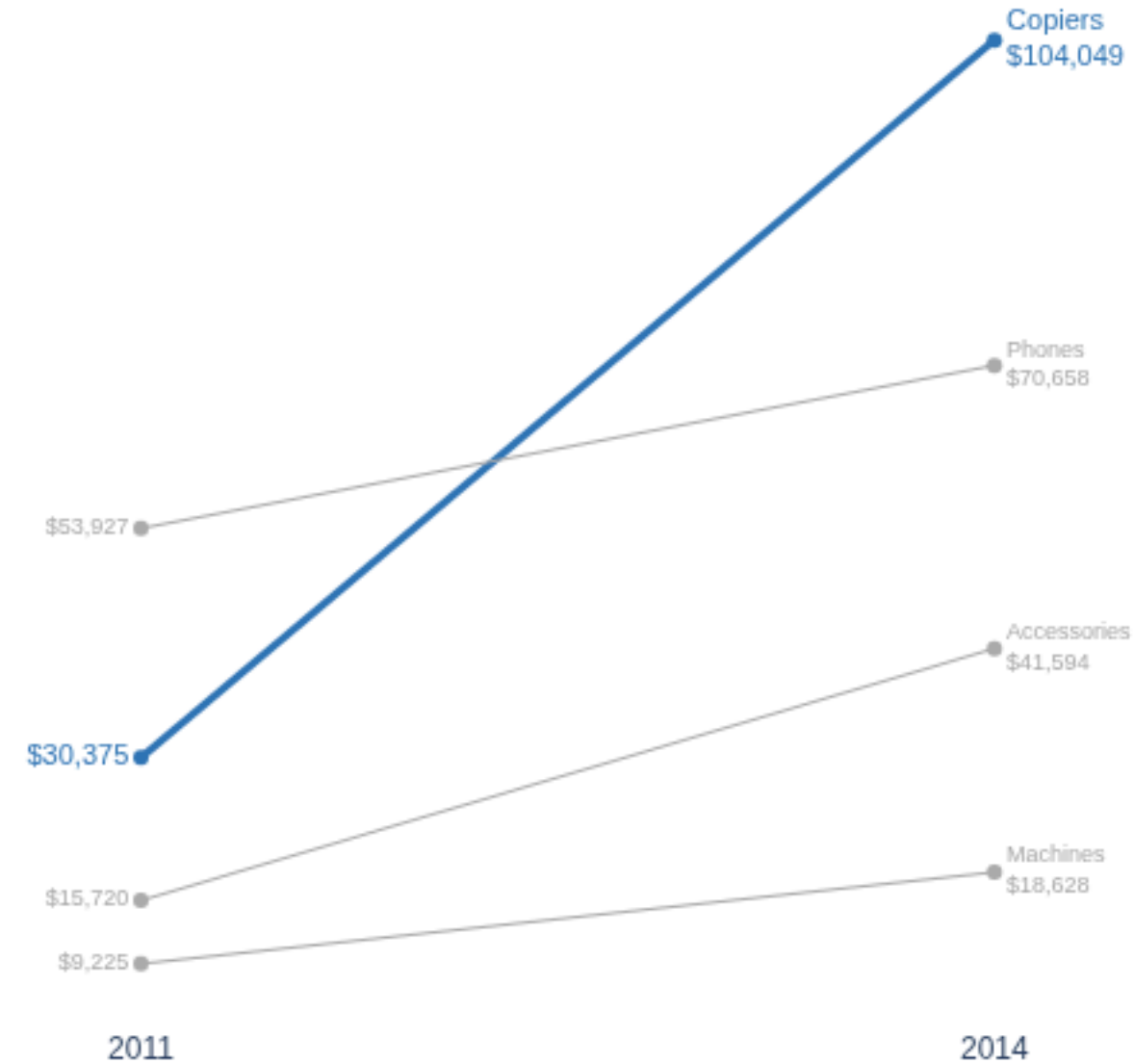
Chart type:

Slopegraph — two points in time; slope IS the story

Insight:

Technology (the most profitable category overall) is being carried by Copiers' rapid growth — a clear reinvestment candidate.

Copiers profit grew the most within Technology — nearly quadrupling from 2011 to 2014



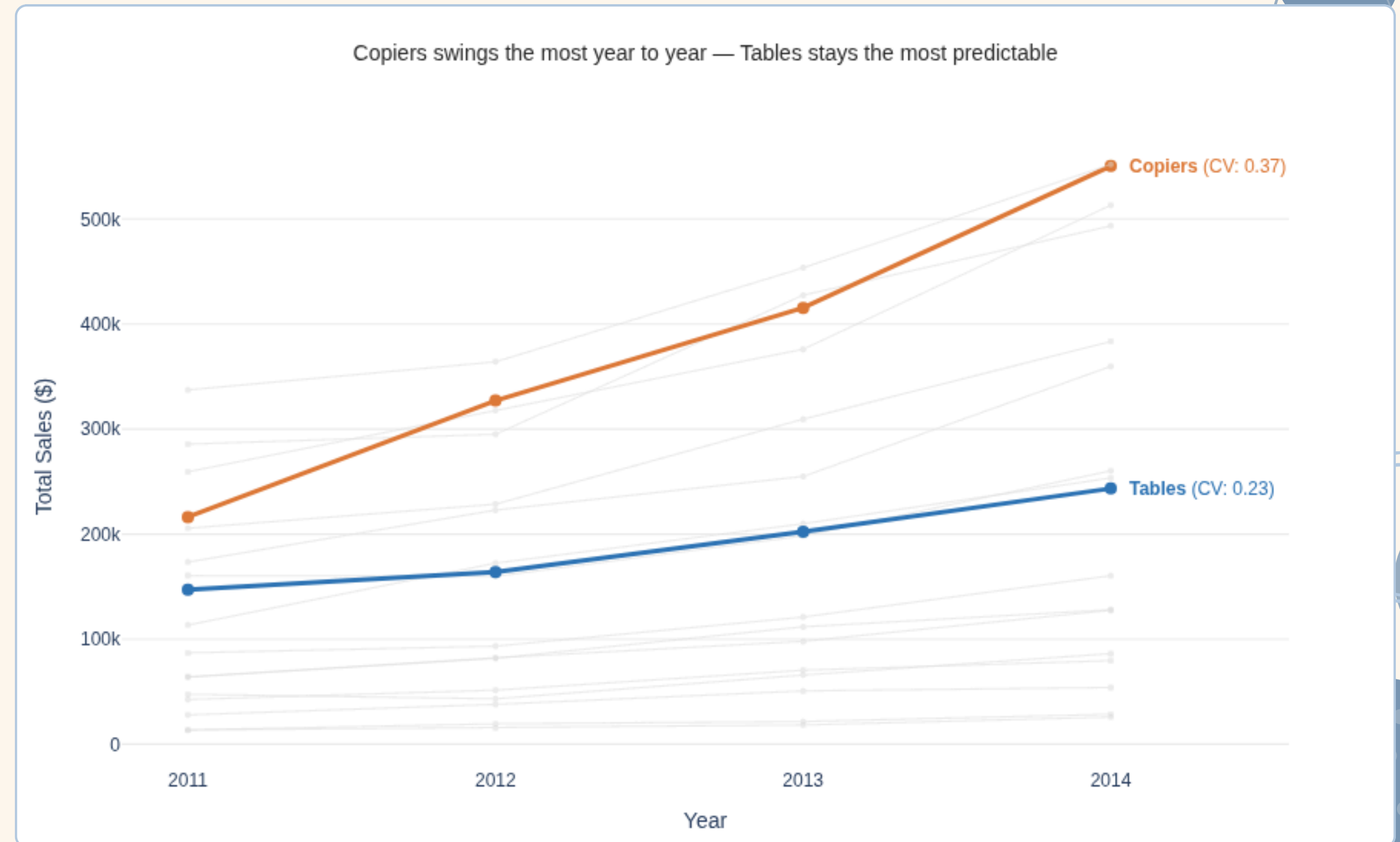
Which product sub-categories show the most volatile sales patterns across years, and which are the most stable?

Chart type:

Multi-line, volatility measured via coefficient of variation.

Insight:

Fast growth (Copiers) comes with volatility — worth pairing growth investment with demand forecasting, not assuming linear growth.



Conclusions & Recommendations

- Cap discounts near 20-25% on Furniture/Tables — beyond this, orders actively lose money.
- Investigate Southeast Asia (Furniture) and Nigeria — isolated but real loss centers worth a root-cause review.
- Double down on Home Office and Copiers — fastest organic growth, worth prioritizing in inventory/marketing.
- Reconsider expedited shipping pricing — Same Day costs 2x to fulfill with zero order-value lift.



Thank you

Dashboard: [[dashboard-url](#)] | Repo: [[github-url](#)] | Contact: [matale.aditya@gmail.com]